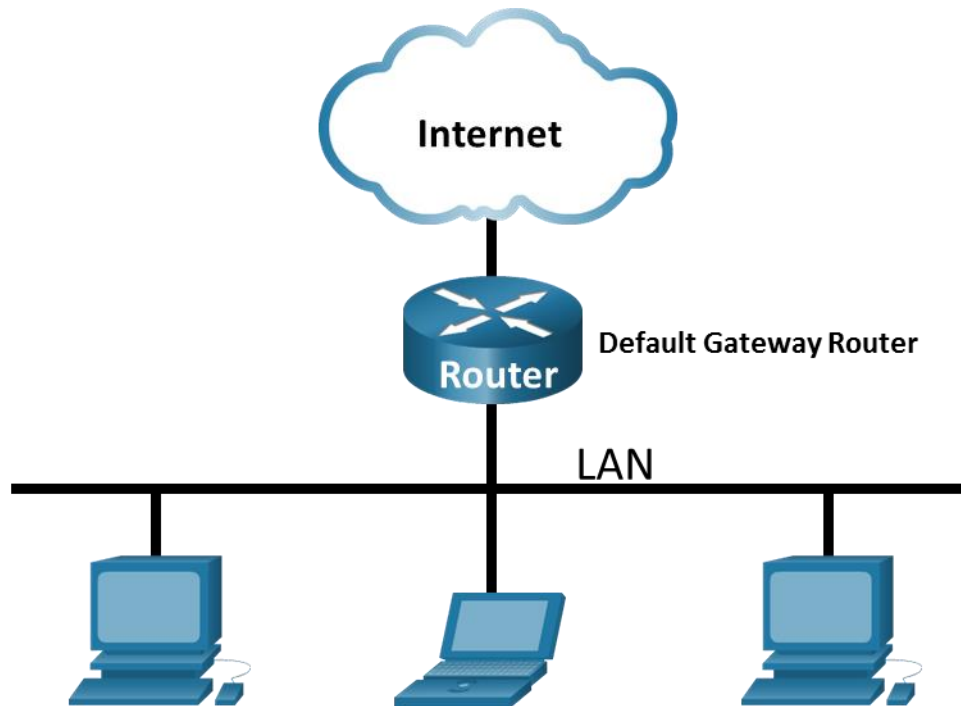


Lab - Use Wireshark to View Network Traffic

Topology



Objectives

Part 1: Capture and Analyze Local ICMP Data in Wireshark

Part 2: Capture and Analyze Remote ICMP Data in Wireshark

Background / Scenario

Wireshark is a software protocol analyzer, or "packet sniffer" application, used for network troubleshooting, analysis, software and protocol development, and education. As data streams travel back and forth over the network, the sniffer "captures" each protocol data unit (PDU) and can decode and analyze its content according to the appropriate RFC or other specifications.

Wireshark is a useful tool for anyone working with networks and can be used with most labs in the CCNA courses for data analysis and troubleshooting. In this lab, you will use Wireshark to capture ICMP data packet IP addresses and Ethernet frame MAC addresses.

Required Resources

- 1 PC (Windows with internet access)
- Additional PCs on a local-area network (LAN) will be used to reply to ping requests.

Using a packet sniffer such as Wireshark may be considered a breach of the security policy of the school. It is recommended that permission be obtained before running Wireshark for this lab. If using a packet sniffer such as Wireshark is an issue, the instructor may wish to assign the lab as homework or perform a walk-through demonstration.

Instructions

Part 1: Capture and Analyze Local ICMP Data in Wireshark

In Part 1 of this lab, you will ping another PC on the LAN and capture ICMP requests and replies in Wireshark. You will also look inside the frames captured for specific information. This analysis should help to clarify how packet headers are used to transport data to their destination.

Step 1: Retrieve your PC interface addresses.

For this lab, you will need to retrieve your PC IP address and its network interface card (NIC) physical address, also called the MAC address.

- a. In a command prompt window, enter **ipconfig /all**, to the IP address of your PC interface, its description, and its MAC (physical) address.

```
C:\Users\Student> ipconfig /all
```

```
Windows IP Configuration
```

```
Host Name . . . . . : DESKTOP-NB48BTC
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

```
Ethernet adapter Ethernet:
```

```
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) 82577LM Gigabit Network Connection
Physical Address. . . . . : 00-26-B9-DD-00-91
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::d809:d939:110f:1b7f%20 (Preferred)
IPv4 Address. . . . . : 192.168.1.147 (Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
```

```
<output omitted>
```

- b. Ask a team member or team members for their PC IP address and provide your PC IP address to them. Do not provide them with your MAC address at this time.

Step 2: Start Wireshark and begin capturing data.

- a. Navigate to Wireshark. Double-click the desired interface to start the packet capture. Make sure the desired interface has traffic.
- b. Information will start scrolling down the top section in Wireshark. The data lines will appear in different colors based on protocol.

This information can scroll by very quickly depending on what communication is taking place between your PC and the LAN. We can apply a filter to make it easier to view and work with the data that is being captured by Wireshark.

Lab - Use Wireshark to View Network Traffic

For this lab, we are only interested in displaying ICMP (ping) PDUs. Type **icmp** in the **Filter** box at the top of Wireshark and press **Enter**, or click the **Apply** button (arrow sign) to view only ICMP (ping) PDUs.

- c. This filter causes all data in the top window to disappear, but you are still capturing the traffic on the interface. Navigate to a command prompt window and ping the IP address that you received from your team member.

```
C:\> ping 192.168.1.114
```

Pinging 192.168.1.114 with 32 bytes of data:

Reply from 192.168.1.114: bytes=32 time<1ms TTL=128

Reply from 192.168.1.114: bytes=32 time<1ms TTL=128

Reply from 192.168.1.114: bytes=32 time<1ms TTL=128

Reply from 192.168.1.114: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.114:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

Notice that you start seeing data appear in the top window of Wireshark again.

The screenshot shows the Wireshark interface with the filter 'icmp' applied. The packet list shows 10 ICMP Echo (ping) request and reply packets. The packet details pane shows the structure of an Ethernet II frame, including the destination and source MAC addresses. The packet bytes pane shows the raw data of the selected packet, which is an ICMP Echo request.

No.	Time	Source	Destination	Protocol	Length	Info
3	10.696465	192.168.1.147	192.168.1.114	ICMP	74	Echo (ping) request id=0x0001, ...
4	10.781036	192.168.1.114	192.168.1.147	ICMP	74	Echo (ping) reply id=0x0001, ...
5	11.718986	192.168.1.147	192.168.1.114	ICMP	74	Echo (ping) request id=0x0001, ...
6	11.805097	192.168.1.114	192.168.1.147	ICMP	74	Echo (ping) reply id=0x0001, ...
7	12.734584	192.168.1.147	192.168.1.114	ICMP	74	Echo (ping) request id=0x0001, ...
8	12.829155	192.168.1.114	192.168.1.147	ICMP	74	Echo (ping) reply id=0x0001, ...
9	13.750216	192.168.1.147	192.168.1.114	ICMP	74	Echo (ping) request id=0x0001, ...
10	13.853254	192.168.1.114	192.168.1.147	ICMP	74	Echo (ping) reply id=0x0001, ...

> Frame 3: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
▼ Ethernet II, Src: Dell_dd:00:91 (00:26:b9:dd:00:91), Dst: Apple_1e:80:72 (28:37:37:1e:80:72)
 ▼ Destination: Apple_1e:80:72 (28:37:37:1e:80:72)
 Address: Apple_1e:80:72 (28:37:37:1e:80:72)
 = LG bit: Globally unique address (factory default)
 = IG bit: Individual address (unicast)
 ▼ Source: Dell_dd:00:91 (00:26:b9:dd:00:91)

0000 28 37 37 1e 80 72 00 26 b9 dd 00 91 08 00 45 00 (77..r.&E.
0010 00 3c 0e 61 00 00 80 01 00 00 c0 a8 01 93 c0 a8 .<.a.... ..
0020 01 72 08 00 4d 4a 00 01 00 11 61 62 63 64 65 66 .r..MJ.. ..abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 wabcdfg hi

Specifies if this is an individual (unicast) o...broadcast/multicast address (eth.ig), 3 byte: Packets: 10 · Displayed: 8 (80.0%) Profile: Default

No.	Time	Source	Destination	Protocol	Length	Info
854	80.080500100	192.168.1.190	192.168.1.246	ICMP	74	Echo (ping) request id=0x0001, seq=23/5888, ttl=128 (reply in 855)
855	80.093674500	192.168.1.246	192.168.1.190	ICMP	74	Echo (ping) reply id=0x0001, seq=23/5888, ttl=128 (request in 854)
863	81.101759700	192.168.1.190	192.168.1.246	ICMP	74	Echo (ping) request id=0x0001, seq=24/6144, ttl=128 (reply in 864)
864	81.105710300	192.168.1.246	192.168.1.190	ICMP	74	Echo (ping) reply id=0x0001, seq=24/6144, ttl=128 (request in 863)
867	82.113774700	192.168.1.190	192.168.1.246	ICMP	74	Echo (ping) request id=0x0001, seq=25/6400, ttl=128 (reply in 868)
868	82.118861200	192.168.1.246	192.168.1.190	ICMP	74	Echo (ping) reply id=0x0001, seq=25/6400, ttl=128 (request in 867)
874	83.124093500	192.168.1.190	192.168.1.246	ICMP	74	Echo (ping) request id=0x0001, seq=26/6656, ttl=128 (reply in 875)
875	83.127652400	192.168.1.246	192.168.1.190	ICMP	74	Echo (ping) reply id=0x0001, seq=26/6656, ttl=128 (request in 874)

Note: If the PC of your team member does not reply to your pings, this may be because the PC firewall of the team member is blocking these requests. Please see Appendix A: Allowing ICMP Traffic Through a Firewall for information on how to allow ICMP traffic through the firewall using Windows.

- d. Stop capturing data by clicking the **Stop Capture** icon.

Step 3: Examine the captured data.

In Step 3, examine the data that was generated by the ping requests of your team member PC. Wireshark data is displayed in three sections: 1) The top section displays the list of PDU frames captured with a summary of the IP packet information listed; 2) the middle section lists PDU information for the frame selected in the top part of the screen and separates a captured PDU frame by its protocol layers; and 3) the bottom section displays the raw data of each layer. The raw data is displayed in both hexadecimal and decimal form.

- Click the first ICMP request PDU frames in the top section of Wireshark. Notice that the **Source** column has your PC IP address, and the **Destination** column contains the IP address of the teammate PC that you pinged.
- With this PDU frame still selected in the top section, navigate to the middle section. Click the plus sign to the left of the Ethernet II row to view the destination and source MAC addresses.

Does the source MAC address match your PC interface?

yes!

Does the destination MAC address in Wireshark match your team member MAC address?

yes!

How is the MAC address of the pinged PC obtained by your PC?

When the PC is pinged, it uses ARP, which is the Address Resolution Protocol. It maps between the hardware addresses and the IPv4 addresses. (3.3)

Note: In the preceding example of a captured ICMP request, ICMP data is encapsulated inside an IPv4 packet PDU (IPv4 header) which is then encapsulated in an Ethernet II frame PDU (Ethernet II header) for transmission on the LAN.

Part 2: Capture and Analyze Remote ICMP Data in Wireshark

In Part 2, you will ping remote hosts (hosts not on the LAN) and examine the generated data from those pings. You will then determine what is different about this data from the data examined in Part 1.

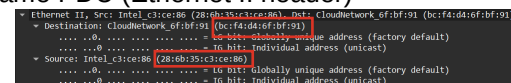
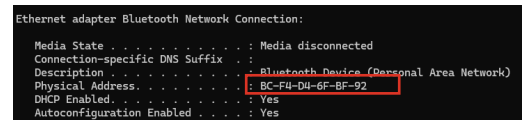
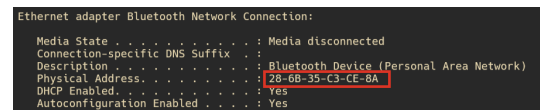
Step 1: Start capturing data on the interface.

- Start the data capture again.
- A window prompts you to save the previously captured data before starting another capture. It is not necessary to save this data. Click **Continue without Saving**.
- With the capture active, ping the following three website URLs from a Windows command prompt:

- 1) www.yahoo.com
- 2) www.cisco.com
- 3) www.google.com

Note: When you ping the URLs listed, notice that the Domain Name Server (DNS) translates the URL to an IP address. Note the IP address received for each URL.

- d. You can stop capturing data by clicking the **Stop Capture** icon.



Step 2: Examining and analyzing the data from the remote hosts.

Review the captured data in Wireshark and examine the IP and MAC addresses of the three locations that you pinged. List the destination IP and MAC addresses for all three locations in the space provided.

IP address for **www.yahoo.com**:

2001:4998:1c:800::1000

MAC address for **www.yahoo.com**:

04:a2:22:f6:f1:09

IP address for **www.cisco.com**:

2600:1401:d000:58c::b33

MAC address for **www.cisco.com**:

04:a2:22:f6:f1:09

IP address for **www.google.com**:

2001:4860:4826:7700::

MAC address for **www.google.com**:

04:a2:22:f6:f1:09

What is significant about this information?

all the mac addresses are the same.

How does this information differ from the local ping information you received in Part 1?

Instead of accessing the ip address directly like we did in part 1, we are not in our local network anymore, and we need to use a default gateway or router in order to connect with these sites.

Reflection Question

Why does Wireshark show the actual MAC address of the local hosts, but not the actual MAC address for the remote hosts?

We don't have access for the mac addresses for the remote hosts, and there might be several of them depending on how large the company is (and multiple ips). This is probably why the mac addresses are the same for all three, it is using the address of the router. We do have the address of our local hosts because we are on the same network.

Appendix A: Allowing ICMP Traffic Through a Firewall

If the members of your team are unable to ping your PC, the firewall may be blocking those requests. This appendix describes how to create a rule in the firewall to allow ping requests. It also describes how to disable the new ICMP rule after you have completed the lab.

Part 1: Create a new inbound rule allowing ICMP traffic through the firewall.

- Navigate to the **Control Panel** and click the **System and Security** option in the Category view.
- In the **System and Security** window, click **Windows Defender Firewall** or **Windows Firewall**.

- c. In the left pane of the **Windows Defender Firewall** or **Windows Firewall** window, click **Advanced settings**.
- d. On the **Advanced Security** window, click the **Inbound Rules** option on the left sidebar and then click **New Rule...** on the right sidebar.
- e. This launches the **New Inbound Rule** wizard. On the **Rule Type** screen, click the **Custom** radio button and click **Next**.
- f. In the left pane, click the **Protocol and Ports** option and using the **Protocol Type** drop-down menu, select **ICMPv4**, and then click **Next**.
- g. Verify that **Any IP address** for both the local and remote IP addresses are selected. Click **Next** to continue.
- h. Select **Allow the connection**. Click **Next** to continue.
- i. By default, this rule applies to all the profiles. Click **Next** to continue.
- j. Name the rule with **Allow ICMP Requests**. Click **Finish** to continue. This new rule should allow your team members to receive ping replies from your PC.

Part 2: Disabling or deleting the new ICMP rule.

After the lab is complete, you may want to disable or even delete the new rule you created in Step 1. Using the **Disable Rule** option allows you to enable the rule again at a later date. Deleting the rule permanently deletes it from the list of inbound rules.

- a. On the **Advanced Security** window, click **Inbound Rules** in the left pane and then locate the rule you created previously.

Right-click the ICMP rule and select **Disable Rule** if so desired. You may also select **Delete** if you want to permanently delete it. If you choose this option, you must re-create the rule again to allow ICMP replies.